GONZALO VOLANTE

SOFTWARE SYSTEMS & APPLIED AI ENGINEER | SOLUTIONS ARCHITECT

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PROFESSIONAL SUMMARY

Software Systems and Applied AI Engineer specialized in designing mission-critical platforms, data ingestion pipelines, and process automation architectures. Proven production track record architecting healthcare ERPs and industrial renewable energy management platforms handling over 1.4 million clinical records and real-time IoT hardware telemetry. Strong expertise in LLM multimodal integration (Google Gemini Vision API), data parsing from semi-structured formats (PDFs, utility bills, financial statements), in-memory caching mechanisms, and automated deployment pipelines on Linux environments.

CORE COMPETENCIES

Architecture & Data:
Domain-Driven Design, High-concurrency SQLite (synchronous better-sqlite3), in-memory RAM caching (vrCache 0 ms latency), massive data migration and sanitization (+1.4M clinical records).

Full-Stack & Applied AI:
React, TypeScript, Tailwind CSS, Python (Flask), Google Gemini Vision API (multimodal invoice OCR), deterministic FIFO financial clearing algorithms, REST APIs & Webhooks.

Infrastructure & Automation:
Linux VPS (systemd, Nginx, Gunicorn, Certbot SSL), Playwright RPA (headless bot-bypass), solar telemetry integration (Growatt API), pdfplumber document parsing, GitHub Actions CI/CD.

PROFESSIONAL EXPERIENCE (ACTIVE PRODUCTION SYSTEMS)

Laboratorio Bioquímico SARESA — 2024 – Present · Salta, Argentina

Lead Solutions Architect & Full-Stack Systems Developer | LIS SARESA V4 (Core Laboratory Information System & Medical ERP)

- Architected and deployed LIS SARESA V4, an end-to-end mission-critical laboratory information system and medical ERP serving active healthcare operations.
- Engineered the migration, indexing, and sanitization of +1,400,000 clinical analysis results, +86,000 medical orders, and a registry of +30,000 active patients.
- Implemented an in-memory RAM caching engine (vrCache) that eliminated N+1 database queries, achieving 0 ms latency across 1,000+ age-and-sex-dependent biological reference evaluations.
- Integrated Google Gemini Vision API for automated multimodal extraction of medical reagent supplier invoices and packing slips into validated database records.
- Developed a deterministic FIFO transactional accounting algorithm to automate debt allocation and ledger consolidation across medical supplier accounts.
- Configured automated deployment pipelines on a dedicated Linux VPS utilizing systemd services, synchronous SQLite access, and Nginx with Certbot SSL renewal.

Nuevas Energías (Renewable Energy & Solar Engineering) — Sep 2024 – Present · Salta, Argentina

IT Specialist & Software Solutions Developer | Central Operations Platform & Solar Energy Balance Engine

- Designed and put into production the central internal operating platform, replacing fragmented legacy spreadsheets with a unified web-based management system.
- Programmed automated solar photovoltaic sizing algorithms (On-Grid and Off-Grid), reducing proposal turnaround time from 48 hours to under 3 minutes (-90% cycle reduction).
- Built an automated ingestion and document parsing pipeline using Python (pdfplumber) to analyze provincial electricity bills (EDESA) and detect contracted power transgressions.
- Integrated solar inverter IoT hardware telemetry (Growatt API) with utility billing data in an isolated SQLite database (plantas.db) for automated net-metering energy balances.
- Built a custom read-only CRM client with local persistent caching to interface with external APIs, eliminating rate-limit penalties while maintaining real-time commercial tracking.
- Implemented full CI/CD deployment automation using GitHub Actions, compiling React/TypeScript frontends and syncing to a production Linux host running Gunicorn and Nginx.

SPECIALIZED ENGINEERING PROJECTS

Publi-Prop — `Headless Web Automation & Browser Session Orchestrator (Node.js, Playwright, Chromium, Session Storage)`. Built an automated real estate publishing pipeline capable of bypassing complex web forms through headless Chromium execution and persistent session management.

Auditor EDESA — `Electrical Tariff Ingestion & Analytical Engine (Python 3.11, pdfplumber, SQLite, Flask, WeasyPrint)`. Analytical document parser extracting time-differentiated hourly energy consumption patterns and modeling mathematical penalty curves into structured audit reports.

OTBN Geospatial Platform — `Satellite Remote Sensing & GIS (Python, Google Earth Engine API, QGIS, GeoPandas)`. Spatial analysis scripts supporting the Technical Committee for the Native Forests Law (Ley de Bosques) in Salta, extracting NDVI and NDWI vegetation indices over Sentinel and Landsat imagery.

Notion Statement Importer — `Standalone Financial Parsing Engine (Python, Notion API, PyInstaller, REST APIs)`. Standalone executable ingesting credit card PDF statements with automatic tagging, duplicate filtering, and REST API database synchronization.

COLLABORATIVE ENGINEERING & SOFTWARE FACTORY

YouCreate 2.0 (iGrowker, 2024) — Frontend Lead & Squad Contributor

Led frontend engineering squad delivering modular financial balance dashboards using React, TypeScript, and Tailwind CSS.

Smart Projects (Foo Talent Group, 2024) — Software Developer

Agile team simulation with continuous Git/GitHub branch management, code reviews, and sprint delivery cadences.

EDUCATION & CERTIFICATIONS

Certified Tech Developer — Software Engineering Program | Digital House (2021 – 2023)

UX/UI Design Specialization | Coderhouse (2021)

LANGUAGES

Spanish: Native | English: B2 Professional (Advanced technical reading, architecture documentation, and fluent communication across international engineering teams).